Quiz 1: Practice

15 minutes 10 points

Name: _____ **UW NetID:** _____ **Campus ID:** _____

Final answers **For TA:** Total _____ / 10

1(a)	1(b)	1(c)	2(a)	2(b)

Lectures 1–3. Not graded. Closed book, with the formulas below. Work individually; no personal notes or AI tools. **Write final answers in the table above:** numbers for 1(a)–(c), one letter (A–D) for each of 2(a)–(b). Exact fractions or decimals are accepted. No partial credit. Try the questions before reading the solutions.

Reference: Bellman expectation equations. For a deterministic policy π:

$$V^\pi(s) = r(s, \pi(s)) + \gamma \sum_{s'} P(s' \mid s, \pi(s)) V^\pi(s'),$$

$$Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^\pi(s').$$

1. State and action values

6 points

For (a)–(b): A machine has states G (good) and W (worn). Use $\gamma = \frac{1}{2}$ and $\pi(\mathrm{G}) = \pi(\mathrm{W}) = \texttt{run}$. The table gives the rewards and transitions for this policy. Each step pays the listed reward, then moves to the next state. The process continues indefinitely.

State s	Action a	$r(s, a)$	$P(\mathrm{G} \mid s, a)$	$P(\mathrm{W} \mid s, a)$
G	run	4	1/2	1/2
W	run	1	0	1

(a) [2 points] Compute $V^\pi(\mathrm{W})$.

(b) [2 points] Compute $V^\pi(\mathrm{G})$.

(c) [2 points] In a separate MDP, use $\gamma = \frac{1}{2}$. Under a fixed policy π, $V^\pi(\mathrm{Y}) = 10$. Taking action a in state X pays -2 and moves to Y with probability 1. Compute $Q^\pi(\mathrm{X}, a)$.

2. Concepts

4 points

(a) [2 points] A website observes each arriving visitor's attributes, chooses one ad, and sees a click or no click only for that ad. Click probabilities are unknown. Future visitors' attributes and responses are unaffected by previous ad choices. What is the most specific formulation?

A. Supervised learning.
B. Contextual bandit.
C. Planning with a known model.
D. Multi-step reinforcement learning with action-dependent future states.

(b) [2 points] A store uses current inventory as its state. Two histories have the same inventory and the same new order, but one has a shipment already in transit and the other does not. Their next-inventory distributions differ. Which conclusion follows?

A. Inventory is Markov because the current action is the same.
B. The two histories show that the ordering policy must be stochastic.
C. Inventory is not Markov; the state must also distinguish pending shipments.
D. No Markov state can represent this store.